# Danielle Marques Simões

Aeronautics Institute of Technology (ITA), São José dos Campos, SP, Brazil
São Paulo, SP, Brazil • danielle@ita.br • +55 (11) 95156-0361 • ORCID 0009-0000-6576-7147

## Education

**Aeronautics Institute of Technology (ITA)**, São José dos Campos, SP, Brazil *Expected Mar 2027*
M.S. in Infrastructure

- Thesis: *Influence of calcination temperature on the microstructure of a laboratory-produced one-part geopolymer.* Advisor: Dr. João Claudio Bassan de Moraes
- Full tuition waiver and CAPES graduate research scholarship (2025 – Present)

**Getúlio Vargas Foundation (FGV)**, São Paulo, SP *2022*
Specialization in Project Management

**University of São Paulo (USP)**, São Paulo, SP *2021*
Non-degree graduate student, Master's program in Archaeology (Archaeology in the Provincial Roman World)

**Federal Institute of São Paulo (IFSP)**, São Paulo, SP *2020*
Bachelor of Science in Civil Engineering

## Research Experience

**Aeronautics Institute of Technology (ITA)**, São José dos Campos, SP *2025 – Present*
*Graduate Researcher, CAPES Fellow; Advisor: Dr. João Claudio Bassan de Moraes*
My research focuses on low-carbon binders for infrastructure, in particular geopolymers and alkali-activated materials, and on how the chemistry of the activator and the availability of calcium govern the mechanical performance of these systems.

- Developing one-part ("just add water") geopolymer binders from calcined kaolinitic clay and solid activators, which avoid the handling of corrosive alkaline solutions required by conventional two-part geopolymers. The binder reached a compressive strength of 59 MPa at 28 days, and this work is the basis of a patent application in progress.
- Investigated the effect of calcium on potassium-based metakaolin geopolymers, comparing the compressive strength of systems activated by KOH and by $K_2CO_3$ + $Ca(OH)_2$ (accepted, 3rd International Congress Women in STEAM, 2026).
- Ongoing: effect of the calcination temperature on the binder microstructure; manuscript for *Cement and Concrete Composites* to be submitted after the patent filing.

**Institute of Aeronautics and Space (IAE) and ITA**, São José dos Campos, SP *Mar 2025 – Aug 2025*
*Team Member, IASC/NASA Asteroid Search Campaign; Advisor: Dr. Odayr Ferreira*

**Federal Institute of São Paulo (IFSP)**, São Paulo, SP *Jul 2019 – Jun 2020*
*Undergraduate Researcher, PIBIC/CNPq Fellow; Advisor: Dr. Roberto Maurício Micalli*

- Study of composite materials for structural applications: analysis and comparison of steel-reinforced and bamboo-reinforced concrete structures for use in bioconstruction.

**Federal Institute of São Paulo (IFSP)**, São Paulo, SP *Oct 2016 – Jun 2018*
*Undergraduate Researcher; Advisor: Dr. Luciano Silva*

- Macroscopic characterization of 200 wood samples focused on applications in civil construction, and creation of the IFSP Xylotheque, a reference collection of wood species.

**Institute for Technological Research (IPT)**, São Paulo, SP *Oct 2016 – Dec 2017*
*Research Intern; Advisor: Dr. Geraldo José Zenid*

- Wood analysis and identification focused on applications in civil construction.

## Publications and Presentations

### Construction Materials

- **Marques, D. S.**, Zucarelli, G., Moraes, J. C. B. (2026) Effect of calcium on the compressive strength of potassium-based metakaolin geopolymers: KOH versus $K_2CO_3$ + $Ca(OH)_2$, *3rd International Congress Women in STEAM*, São José dos Campos (accepted).
- **Marques, D. S.**, Moraes, J. C. B. Development of one-part geopolymers. Manuscript in preparation for *Cement and Concrete Composites*; submission pending patent filing.
- **Marques, D. S.**, Souza, W. S., Micalli, R. M. Use of fiber-reinforced concrete in São Paulo subway tunnels (manuscript in preparation).

### Patents

- Geopolymer product and process with reduced $CO_2$ emissions and low energy consumption. Patent application in progress, Brazil, 2026.

### Talks

- **Marques, D. S.** (2026) Self-healing mechanisms in concrete: lessons from the past for building a smarter and more sustainable future, *SEDICITEC 2026*, Federal Institute of São Paulo (IFSP), October 21, 2026. Invited talk.
- **Marques, D. S.** (2025) Concrete technological control of precast lining rings for the TBM tunnels of the São Paulo Metro, *SEDICITEC 2025*, Federal Institute of São Paulo (IFSP).

### Infrastructure Engineering: Conference Papers

- **Marques, D. S.**, Souza, W. S., Silva, C. C. (2026) Application of network topological analysis to the evaluation of a proposed structuring axis for the East Zone of the São Paulo metro network, *AEAMESP, 32nd Metro-Railway Technology Week*, São Paulo. Finalist, 13th Technology and Development Metro-Railway Award.
- **Marques, D. S.**, Souza, W. S., Marques, F. R. (2026) BIM data governance in public infrastructure contracts: automated verification of IFC models in the São Paulo Metro, *9th International Congress: The BIM Era*, Brasília.
- Souza, W. S., **Marques, D. S.**, Silva, C. C. (2026) Computational modeling of diaphragm walls via soil-structure interaction in Python and Matlab, *22nd COBRAMSEG, Brazilian Congress of Soil Mechanics and Geotechnical Engineering*.
- Souza, W. S., **Marques, D. S.**, Silva, E. J. (2025) INSPECTUS: route optimization tool for rural aerodrome inspection using the Traveling Salesman Problem, *22nd SITRAER, Air Transportation Symposium*, Brasília.

### Books and Book Chapters in Preparation

- **Marques, D. S.** (invited author). Chapter in a book of the *Women in Sustainability* series (*Série Mulheres*), Editora Leader.
- Souza, W. S., **Marques, D. S.** High school physics textbook (in Portuguese).

### Undergraduate Conference Presentations

- **Marques, D. S.**, Souza, W. S., Ferreira, G. J., Kenchian, A. (2017) 10th Boldness Competition for Engineers and Architects, *Scientific Initiation Congress of the Federal Institute of São Paulo*, Bento Gonçalves. Poster.
- **Marques, D. S.**, Kenchian, A. (2017) Housing technical advice for low-income populations, *Scientific Initiation Congress of the Federal Institute of São Paulo*, Sertãozinho. Poster.
- **Marques, D. S.**, Cronchik, L. (2016) Art and Science at School Program, *Scientific Initiation Congress of the Federal Institute of São Paulo*, Catanduva. Talk.

## Teaching and Mentoring

**Aeronautics Institute of Technology (ITA)**, São José dos Campos, SP *Jun 2026*
*Examiner (Rapporteur), Undergraduate Final Project Committee (TG-1), Civil-Aeronautical Engineering*

- *Effects of the dosage and particle size of colloidal nano-silica (CNS) on the properties of cementitious systems.* Student: Manuela de Almeida Gomes Rocha; Advisor: Dr. João Cláudio Bassan de Moraes.

**Bonsucesso Community Prep Course (CCB)**, Guarulhos, SP *Jan 2025 – Present*
*Volunteer Physics Teacher*

- Teaching physics to pre-university students in a community course.

**Federal Institute of São Paulo (IFSP)**, São Paulo, SP *Jun 2019 – Jun 2020*
*Teaching Assistant, Structural Engineering*

**Federal Institute of São Paulo (IFSP)**, São Paulo, SP *Jun 2015 – Dec 2015*
*Teaching Assistant, Art and Science at School Program*

- Physics linked to arts for children.

## Professional Experience

**Consórcio ICC**, São Paulo, SP *Sep 2023 – Present*
*Civil Engineer*

- Construction supervision and concrete technological control of the precast lining rings for the TBM tunnels of the São Paulo Metro Line 2-Green expansion.

**Raízen Power**, São Paulo, SP *Aug 2022 – Sep 2023*
*Renewable Energy Project Manager*

**Ekko Group**, São Paulo, SP *Jun 2021 – Aug 2022*
*Project Coordinator, Real Estate*

**Avanza Engenharia**, São Paulo, SP *Nov 2019 – May 2021*
*Civil Engineer, Civil Engineering Consultancy*

**Internships**
**Votorantim Cimentos**, São Paulo, SP *Aug 2018 – Nov 2019*
*Intern, Production of Aggregates for Concrete*

- Monitoring and management of aggregates mining operations.

**Tallento Engenharia**, São Paulo, SP *Jan 2018 – Jul 2018*
*Intern, Construction Control*

- Application of process improvements in construction execution and quality inspection.

## Scholarships and Awards

- Finalist, 13th Technology and Development Metro-Railway Award (*Prêmio Tecnologia & Desenvolvimento Metroferroviários*), AEAMESP 2026
- Best and Largest Works Award (*Melhores e Maiores Obras*), Instituto de Engenharia, with INGENIA Engenharia, Line 2-Green expansion from Vila Prudente to Penha, São Paulo Metro 2024
- Graduate Research Scholarship, CAPES, and Full Tuition Waiver, ITA 2025 – Present
- Undergraduate Research Fellowship, Institutional Scientific Initiation Scholarship Program (PIBIC), CNPq 2019 – 2020

- Undergraduate Assistant Fellowship, IFSP 2015 – 2017
- Undergraduate Scholarship, Full Tuition Waiver, Brazilian Federal Government, IFSP 2015 – 2020

## Media

- Interview on the production and transport of precast concrete tunnel rings for the São Paulo Metro Line 2-Green expansion. *A produção e o transporte dos anéis dos túneis do metrô*, Brasil Caminhoneiro (video report). Link Jun 2024

## Technical Skills

- **Laboratory:** concrete durability tests; mechanical tests.
- **Material characterization:** XRF, XRD, FTIR, SEM.
- **Programming:** MATLAB, Python, C.
- **Software:** TQS, SAP2000.

## Professional Licensure and Memberships

- Registered Civil Engineer, Regional Council of Engineering and Agronomy of São Paulo (CREA-SP), since 2021
- Brazilian Concrete Institute (IBRACON)
  - Member, Technical Committee CT 101 (Concrete Sustainability), Working Group GT 3 (Concrete Materials and Mix Design)
- American Concrete Institute (ACI), Student Chapter
  - Associate Member, ACI Committee 130: Sustainability of Concrete
- International Union of Laboratories and Experts in Construction Materials, Systems and Structures (RILEM)

## Service and Outreach

**Compassiva**, São Paulo, SP *2015 – Present*
*Volunteer Portuguese Teacher, Refugee Support Project*

**Votorantim Cimentos**, São Paulo, SP *2018 – 2019*
*Vice Leader, Volunteer Committee* *2018 – 2019*
*Volunteer Math Teacher, Community Support Project* *2018 – 2019*

**Ático Advice, Federal Institute of São Paulo (IFSP)**, São Paulo, SP *Jan 2016 – Dec 2017*
*Designer Assistant*

- Development of house designs for low-income populations.

**IFSP Junior Company**, São Paulo, SP *Jan 2016 – Jun 2016*
*Planning Coordinator*

- Planning of the civil engineering activities of the junior company.

**Pentecostal Church God is Love**, São Paulo, SP *2014 – 2016*
*Volunteer Music Teacher, Community Support Project*